MUNICIPAL INFRASTRUCTURE DAMAGE REPORTING APP

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SECTION E

Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	User can submit a new damage report.	High	A new report is created in the system with all submitted details.	To allow citizens to report infrastructure damage.
FR-002	Functional	User can attach photos to a damage report.	High	At least one photo is successfully uploaded and associated with the report.	Visual evidence aids in assessment and prioritization.
FR-003	Functional	User can view a list of previously submitted reports.	Medium	All submitted reports by the user are displayed in a chronological list.	To provide users with a history of their reports.
FR-004	Functional	System automatically captures GPS location when a report is submitted.	High	The GPS coordinates (latitude and longitude) of the device are recorded with the report.	To accurately pinpoint the location of the damage.
FR-005	Functional	User can select the type of infrastructure damaged (e.g., road, streetlight, park bench).	High	A predefined list of infrastructure types is available for selection, and the chosen type	To categorize damages for efficient routing to the correct department.

				is saved with	
NFR-001	Non-Functional	The application must be available 99.5% of the time.	High	Downtime does not exceed 21.9 hours per month.	Ensures citizens can report damages whenever needed.
NFR-002	Non-Functional	Report submission process should take no more than 30 seconds on a stable internet connection.	Medium	Average submission time is measured and confirmed to be under 30 seconds.	To provide a user-friendly and efficient reporting experience.

Use-Case Specification: Report Damage

Use-Case ID: UC-001

Use-Case Name: Report Damage

Preconditions:

- The user has the Municipal Infrastructure Damage Reporting App installed on their device.
- The user has a stable internet connection (for most scenarios).
- The user's device has GPS capabilities enabled.

Postconditions:

- A new damage report is successfully created and stored in the system.
- The report includes details provided by the user, attached media (if any), and location data.

Main Success Scenario:

1. The user opens the Municipal Infrastructure Damage Reporting App.
2. The user taps the 'Report New Damage' button.
3. The app prompts the user to enable location services, which the user permits.
4. The app automatically captures the current GPS coordinates.
5. The user selects the type of infrastructure damaged from a predefined list.
6. The user enters a description of the damage.
7. The user chooses to attach photos and successfully uploads one or more images.

8. The user reviews the report details, including the captured location and attached photos.
9. The user taps the 'Submit' button.
10. The system validates the report data.
11. The system saves the damage report with all associated information.
12. The app displays a confirmation message to the user indicating successful submission.

Alternate Flow: Missing GPS

1. Steps 1-3 of the Main Success Scenario are followed.
2. The app detects that GPS services are unavailable or have been denied by the user.
3. The app prompts the user to manually enter the location or allows them to select a location on a map.
4. The user manually inputs the address or pinpoints the location on the map.
5. The user proceeds with steps 5-12 of the Main Success Scenario, with the manually entered location replacing the GPS coordinates.
6. The system saves the damage report with the manually provided location information.

UML Use-Case Diagram

